4 5 6

Htw 3 Due 10/25

① 1) $E[X] = n \cdot p = 3 \cdot \frac{1}{4} = \frac{3}{4}$

$E[X^2] - E[X] = np + n(n-1)p^2 - (np)$

$= \frac{3}{4} + 3(3-1)\left(\frac{1}{4}\right)^2 - \frac{3}{4} = 3 \cdot 2 \cdot \frac{1}{16} = \frac{3}{8}$

$\sigma^2 = E[X(X-1)] + E[X] - \mu^2 = \frac{3}{8} + \frac{3}{4} - \left(\frac{3}{4}\right)^2 = \frac{3}{8} + \frac{3}{4} - \frac{9}{16} =$

$\frac{6}{16} + \frac{12}{16} - \frac{9}{16} = \frac{9}{16}$

2) $E[X] = n \cdot p = 4 \cdot \frac{1}{2} = 2$

$E[X^2] - E[X] = np + n(n-1)p^2 - (np)$

$= 2 + 4(3)\left(\frac{1}{4}\right) - 2 = 2 + 3 - 2 = 3$

$\sigma^2 = E[X(X-1)] + E[X] - \mu^2 = 3 + 2 - (2^2) = 5 - 4 = 1$

② Mean $= E[X] = (1)\left(\frac{1}{16}\right) + (2)\left(\frac{3}{16}\right) + (3)\left(\frac{5}{16}\right) + (4)\left(\frac{7}{16}\right) =$

$= \frac{1}{16} + \frac{3}{8} + \frac{15}{16} + \frac{14}{8} = \frac{50}{16} = \frac{25}{8}$

Variance $= \sigma^2 = E[X^2] - \mu^2$

$= (1)^2\left(\frac{1}{16}\right) + (4)\left(\frac{3}{16}\right) + (9)\left(\frac{5}{16}\right) + (16)\left(\frac{7}{16}\right) - \left(\frac{25}{8}\right)^2$

$= \frac{1}{16} + \frac{3}{4} + \frac{45}{16} + 7 - \left(\frac{25}{8}\right)^2$

$= \frac{170}{16} - \frac{625}{64} = 10.625 - 9.77 \approx 0.855$

$SD = \sqrt{0.855} = 0.92$

③ $M'(t) = \frac{2}{5}e^t + \frac{2}{5}e^{2t} + \frac{6}{5}e^{3t}$

Mean $M'(0) = \frac{2}{5} + \frac{2}{5} + \frac{6}{5} = 2$

var $\sigma^2 = M''(0) - 2^2 = \frac{2}{5}e^0 + \frac{4}{5}e^{2 \cdot 0} + \frac{18}{5}e^{3 \cdot 0} - 4 = \frac{2}{5} + \frac{4}{5} + \frac{18}{5} - 4 = \frac{24}{5} - 4 =$

$\frac{24}{5} - \frac{20}{5} = \frac{4}{5}$

pmf $P(X=1) = \frac{2}{5}$ $P(X=2) = \frac{1}{5}$ $P(X=3) = \frac{2}{5}$

$$p(x) = \begin{cases} \frac{2}{5}, & \text{if } x = 1 \\ \frac{1}{5}, & \text{if } x = 2 \\ \frac{2}{5}, & \text{if } x = 3 \\ 0, & \text{all other } x \end{cases}$$

④ 1) $\mu = np = 5 \cdot 0.7 = \boxed{3.5}$ $\sigma^2 = np(1-p) = 3.5(0.3) = \boxed{1.05}$

$P(X=x) = \sum_{x=1}^{\infty} \binom{5}{x} 0.7^x (0.3)^{5-x}$

$f(1) + f(2) = \binom{5}{1} 0.7^1 \cdot 0.3^4 + \binom{5}{2} 0.7^2 \cdot 0.3^3 = 0.02835 + 0.1323 = 0.16065$

Mean

2) $M'(t) = \frac{(1-0.7e^t)(0.3e^t) - (0.3e^t)(-0.7e^t)}{(1-0.7e^t)^2}$ $M'(0) = \frac{0.3}{(0.3)^2} = \frac{1}{0.3} = \boxed{\frac{10}{3}}$

$M'(t) = \frac{(0.3e^t)(1-0.7e^t+0.7e^t)}{(1-0.7e^t)^2} = \frac{0.3e^t}{(1-0.7e^t)^2}$

$M''(t) = \frac{(1-0.7e^t)^2(0.3e^t) - [2(-0.7e^t)(1-0.7e^t)(0.3e^t)]}{(1-0.7e^t)^4}$

$M''(0) = \frac{(0.3)^2(0.3) - [2(-0.7)(0.3)(0.3)]}{(0.3)^4} = \frac{0.3^3 + (1.4)(0.3)^2}{} =$

$\frac{0.3+1.4}{0.3^2} = \frac{1.7}{0.09} = \boxed{18.89}$

Var: $18.89 - (\frac{10}{3})^2 \approx \boxed{7.78}$

pmf

$\frac{1}{1-r} = \sum_{i=0}^{\infty} r^i$

$\frac{0.3e^t}{1-0.7e^t} = 0.3e^t\left(\frac{1}{1-0.7e^t}\right) =$

$P(X=x) = 0.3 \times 0.7^{i-1}$

$P(X=1) = 0.3$

$P(X=2) = 0.21$

$P(1 \le x \le 2) = 0.3 + 0.21 = \boxed{0.51}$

$0.3e^t \sum_{i=0}^{\infty} (0.7e^t)^i = 0.3e^t \sum_{i=0}^{\infty} (0.7^i)(e^{ti})$

$= \sum_{i=1}^{\infty} (0.3)(0.7^{i-1})(e^{ti})$

mean var pmf 3) $M'(t) = 0.55e^t$ $M'(0) = \mu = 0.55$ $M''(0) = 0.55e^{(0)} = 0.55$

$M''(0) - [M'(0)]^2 = 0.55 - (0.55)^2 = \boxed{0.2475}$

x can only be 0 or 1 $P(X=1) = 0.55$ $P(X=2) = 0$

$\boxed{P(1 \le x \le 2) = 0.55}$

Mean var 4) $M'(t) = 0.3e^t + 0.8e^{2t} + 0.6e^{3t} + 0.4e^{4t}$

$M'(0) = 0.3 + 0.8 + 0.6 + 0.4 = \boxed{2.1}$ $M''(0) = 3.7$

$M''(t) = 0.3e^t + 1.6e^{2t} + 1.8e^{3t} + 1.6e^{4t}$ $5.3 - (2.1)^2 = \boxed{0.89}$

pmf $P(X=1) + P(X=2) = 0.3 + 0.4 = \boxed{0.7}$

5) $m=10$, using textbook formula: $\mu = \frac{m+1}{2} = \frac{10+1}{2} = \boxed{5.5}$

$\sigma^2 = \frac{10^2-1}{12} = \boxed{\frac{99}{12}}$

pmf $P(X=x) = 0.1$ $x = 1 \ldots 10$

$P(1 \le x \le 2) = P(X=1) + P(X=2) =$

$0.1 + 0.1 = \boxed{0.2}$

p

↓

1/10 win 9/10 lose

⑤ losing tickets = $\frac{45M}{50M}$ approximately independent since 50 M is so large

prob of losing ticket $\approx 45/50$ or $9/10$

1) prob $\geq$ 1 prize $= 1-(1-p)^{10} = 1-(1-\frac{1}{10})^{10} = 1-(\frac{9}{10})^{10} = 0.65$

2) $1-(\frac{9}{10})^{15} = 0.79$

⑥ 5 cards picked

52 total

12 face cards total

This is a hypergeometric random variable since we have
- N_1 obj of type 1 (faces) 12
- N_2 obj of type 2 (all other cards) 40
- picking n objs (5)
- Support $x = \{0, \ldots n \text{ or } 5\}$ since we can get 0, 1, 2, 3, 4, or 5 face cards when we pick 5 random cards

$$f(x) = \frac{\binom{N_1}{x}\binom{N_2}{5-x}}{\binom{52}{5}}$$

$$f(1) = \frac{\binom{12}{1}\binom{40}{4}}{\binom{52}{5}} \approx 0.42$$

$$f(2) = \frac{\binom{12}{2}\binom{40}{3}}{\binom{52}{5}} \approx 0.25$$

$$f(3) = \frac{\binom{12}{3}\binom{40}{2}}{\binom{52}{5}} \approx 0.066$$

$$f(4) = \frac{\binom{12}{4}\binom{40}{1}}{\binom{52}{5}} \approx 0.007$$

$P(1 \leq x \leq 4) \approx$

$0.42 + 0.25 + 0.066 + 0.007$

$= 0.743$